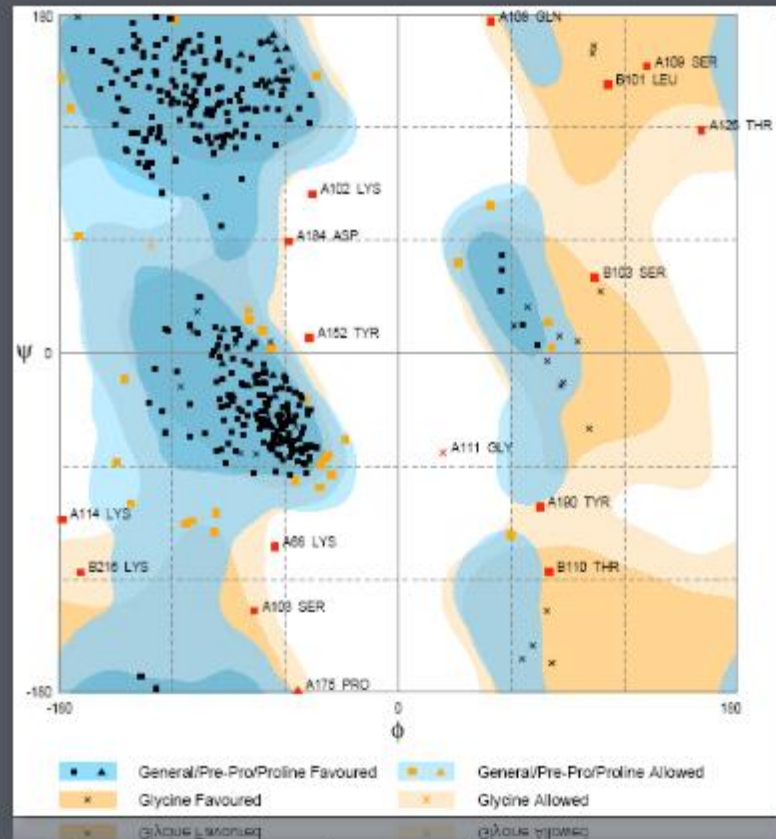
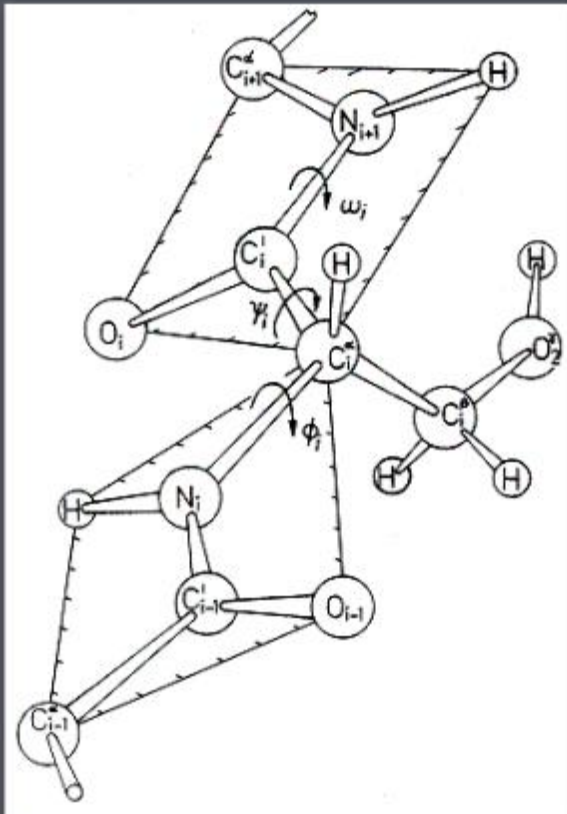


Biophysics (Bio361)

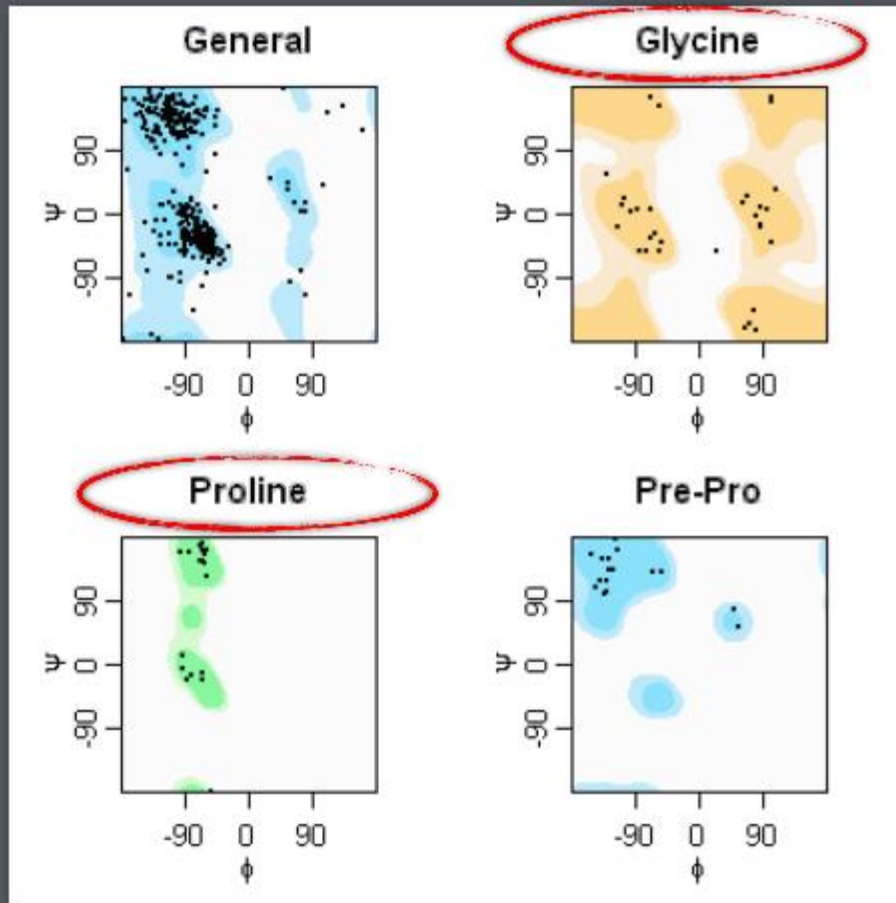
Dr. Arjun Ray
(arjun@iiitd.ac.in)

Ramachandran diagrams

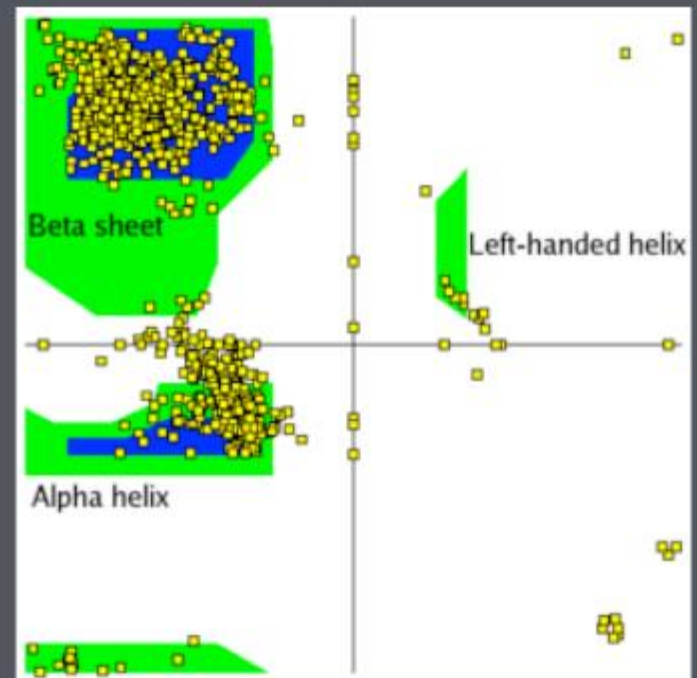


Natural degrees of freedom for amino acids

Ramachandran species

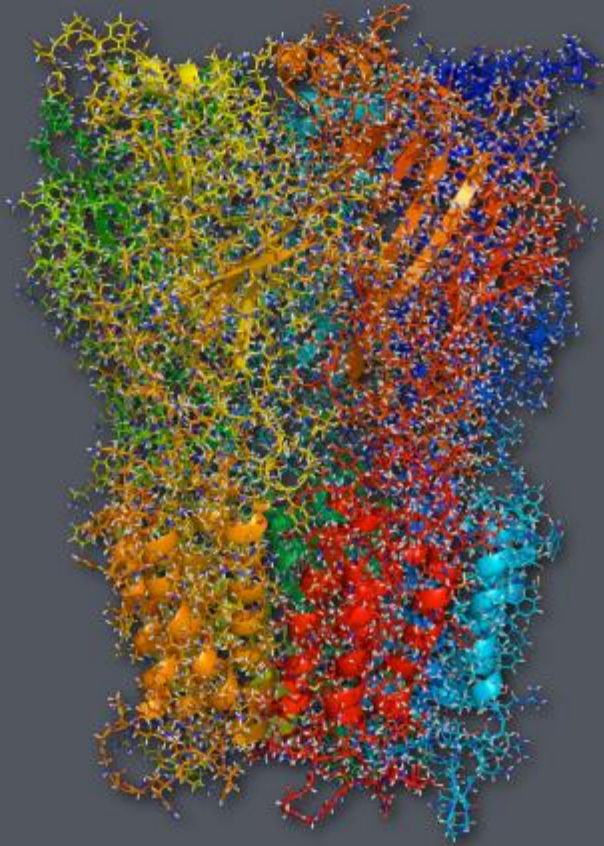


Glycine very flexible (no side chain)
Proline very inflexible; even messes up previous residue!

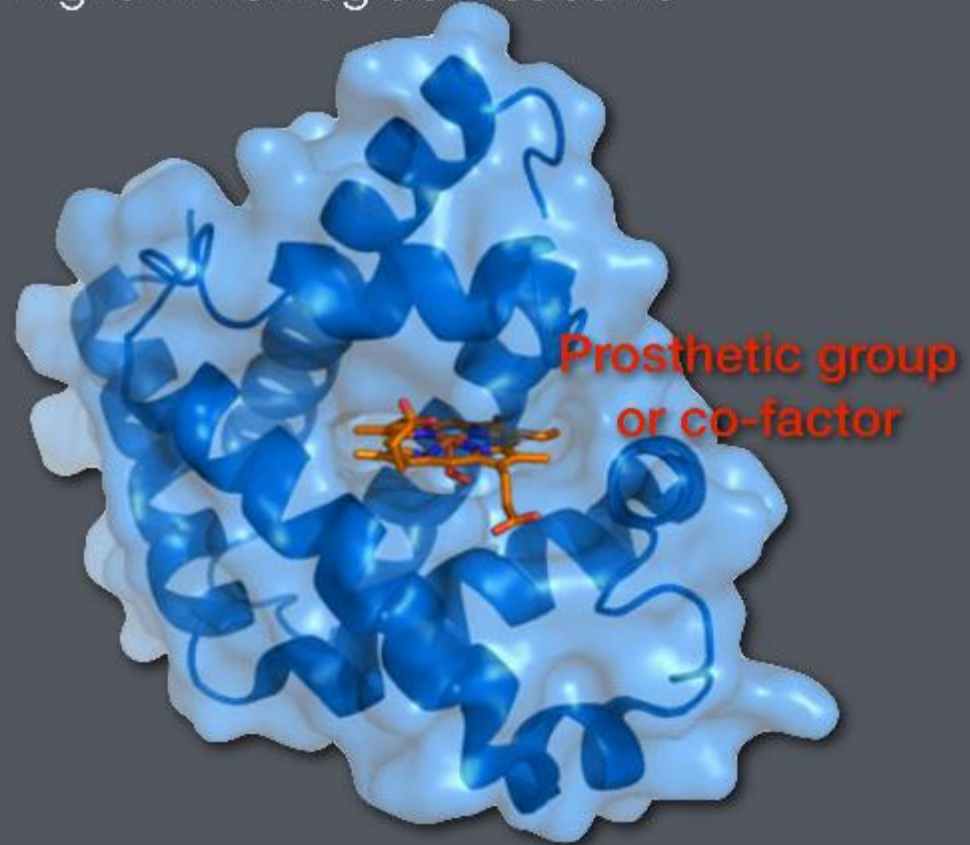


An assembled protein

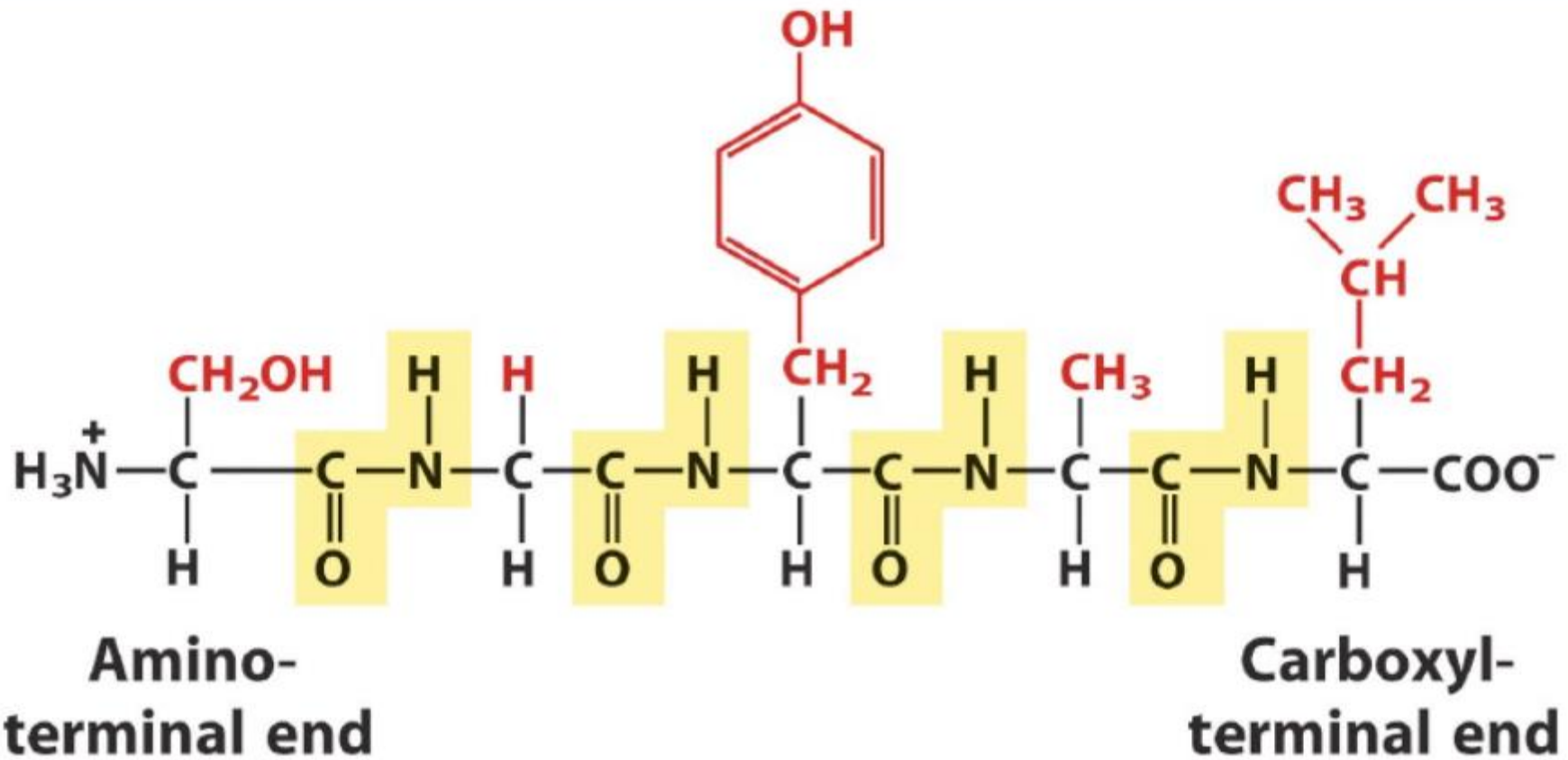
Left: The GLIC ion channel



Right: A hemoglobin subunit



Why this diversity?



Anfinsen & Levinthal

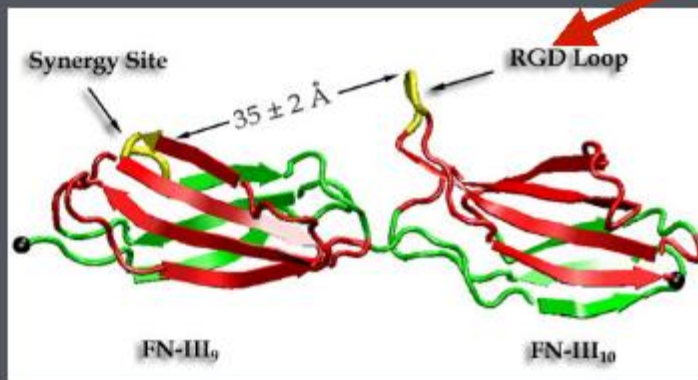
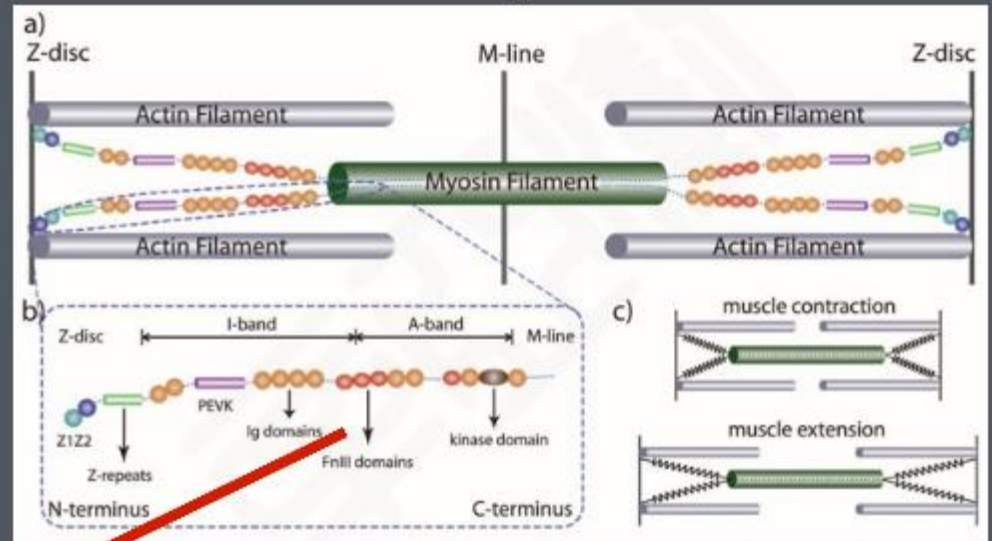
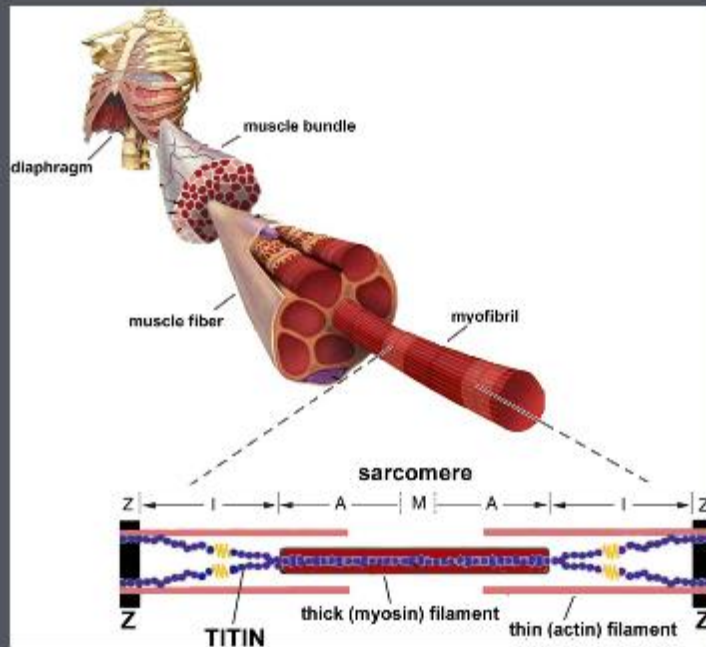
- Christian Anfinsen: Proteins adopt the structure corresponding to the global minimum in free energy (Nobel 1972)
- Cyrus Levinthal: But how can a protein find that? Even with just 2 states per amino acid and 100 residues there are 2^{100} states! (*Levinthal's Paradox*)



Example Proteins

	<i>Molecular weight</i>	<i>Number of residues</i>	<i>Number of polypeptide chains</i>
Cytochrome c (human)	13,000	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (chicken egg white)	13,930	129	1
Myoglobin (equine heart)	16,890	153	1
Chymotrypsin (bovine pancreas)	21,600	241	3
Chymotrypsinogen (bovine)	22,000	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	68,500	609	1
Hexokinase (yeast)	102,000	972	2
RNA polymerase (<i>E. coli</i>)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (<i>E. coli</i>)	619,000	5,628	12
Titin (human)	2,993,000	26,926	1

Case study: Titin



Covered material and suggested reading

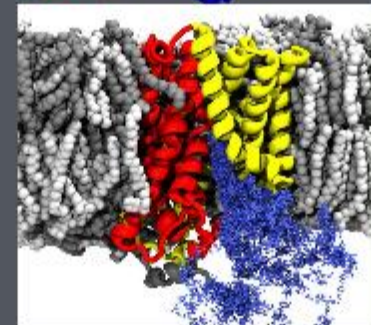
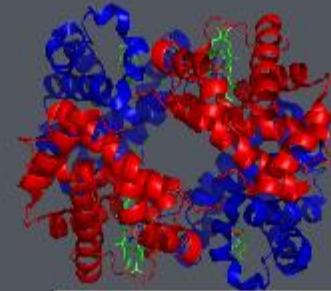
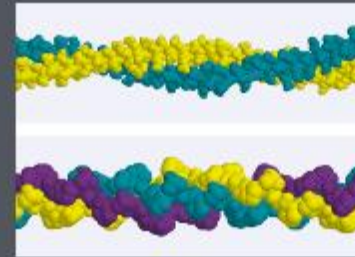
- Finkelstein
 - Chapter 3 & 4
- David Eisenberg - "The discovery of helices and sheets, the principal structural features of proteins", PNAS, 2003
- Cyrus Levinthal - "Molecular Model building by Computer", Scientific American, 1966

Study questions

1. Name three properties by which amino acids can differ.
2. One amino acid is NOT chiral. Which one? Is it special in any way?
3. There is one more highly special amino acid. Which one? How is it special?
4. What would happen if we had some D-isomer amino acids in a protein?
5. How are amino acids linked into a protein chain?
6. What are the levels of structure organization in proteins?
7. Describe the relation between sequence, structure, and function.
8. Does function induce structure?
9. Which ones are the most important degrees of freedom in proteins?
10. Why are cis conformations almost never observed in peptide bonds?
11. What did (1) Anfinsen and (2) Levinthal say?
12. Explain what a Ramachandran plot can tell you about protein geometry.

Protein classification

1. Fibrous proteins
 - Insoluble, strong, highly regular
 - Often form aggregates
 - Lots of hydrogen bonds
2. Globular proteins
 - Water soluble, less regular
 - Complex folds
 - Peptide chain interacts with itself, other domains, and cofactors
3. Membrane Proteins
 - Found in the oily lipid environment
 - Attached to cellular membranes
 - Often channels & transporters



Protein hardness

- Single domain proteins:
 - Really hard, easy to crystallize
 - One compact globule
- Larger proteins:
 - For multidomain proteins, the individual domains are rigid, but the assembly can be flexible (Titin)
- Some proteins change shape slightly as part of their biological function (channels, transporters, binders)
- A solid protein behaves like a crystal: It is firm, and then suddenly melts
- Compare to a light bulb: All-or-nothing, not gradually.

Protein Structure

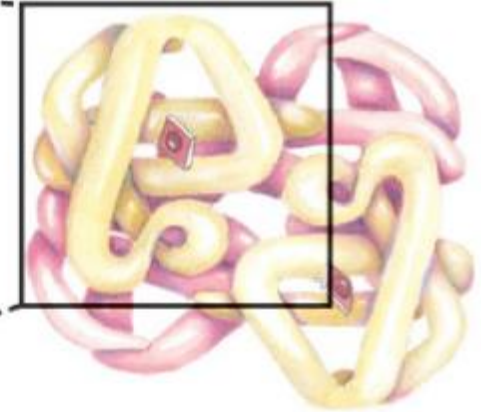
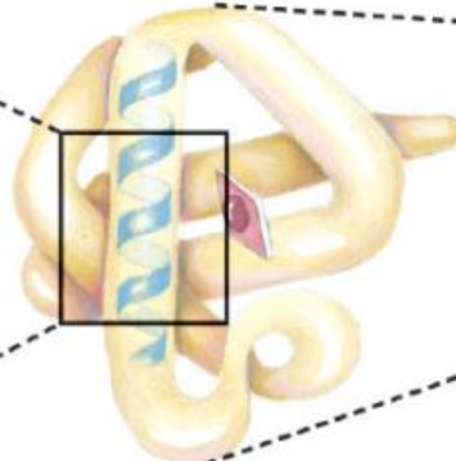
Primary
Structure

Secondary
Structure

Tertiary
Structure

Quaternary
Structure

Lys
Lys
Gly
Gly
Leu
Val
Ala
His



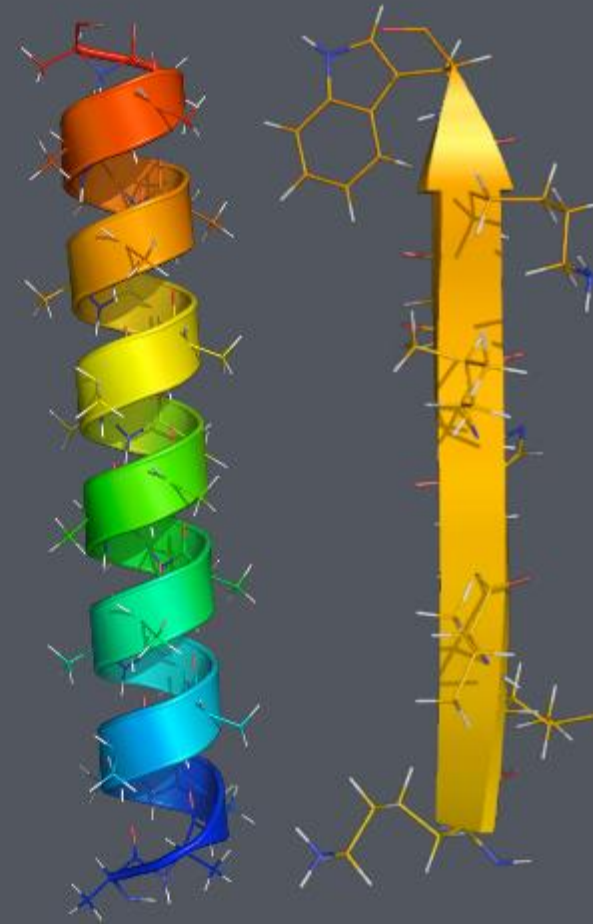
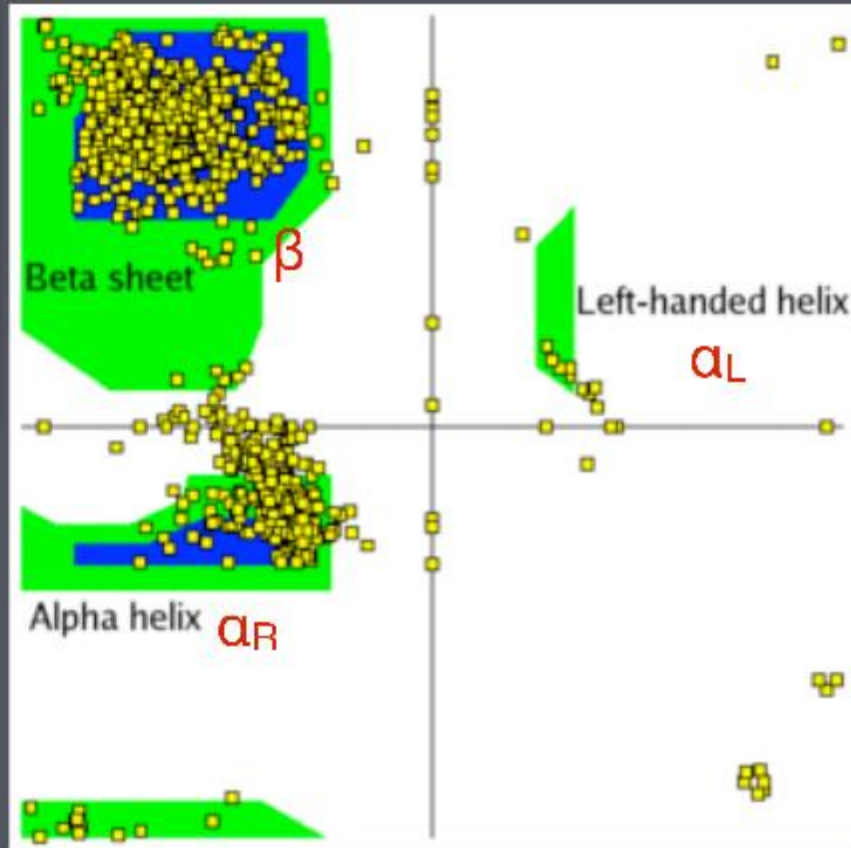
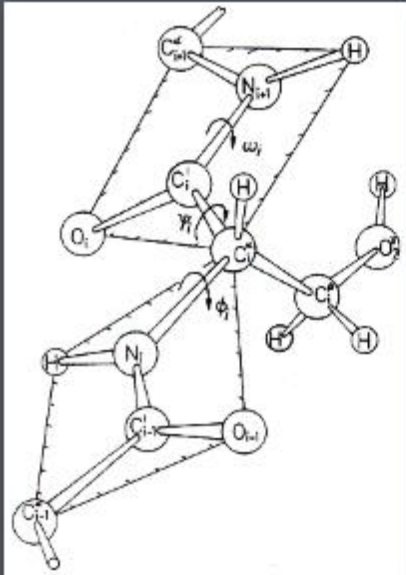
Residues

Helix

Chain

Subunits

Secondary structure



Helices

- Naturally occurring amino acid helices are right-handed
- Nomenclature: N_M -helix
 - Residue i h-bonds to $i+N$
 - M atoms per helical turn
- 3_{10} helix
- 4_{13} (α) helix - most common!
- Other (very rare) forms: 2_7 and 5_{16} (π) helix



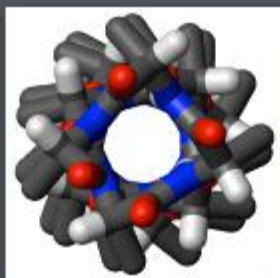
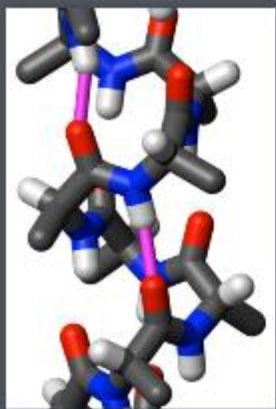
Figure 7-11. The right-handed α -helix.

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Figure 7-11. The right-handed α -helix.

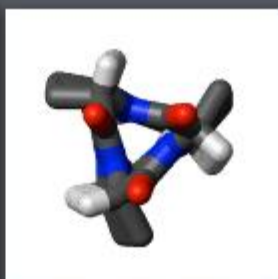
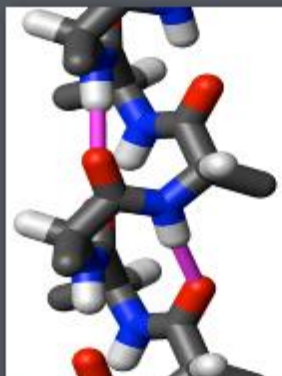
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Helix examples



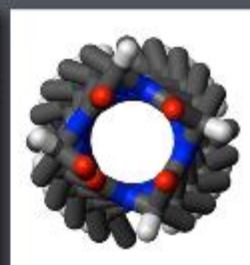
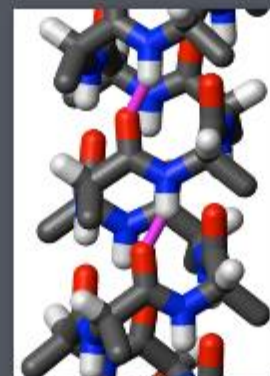
α helix

The α -helix is the most relaxed of the helical structures



3₁₀ helix

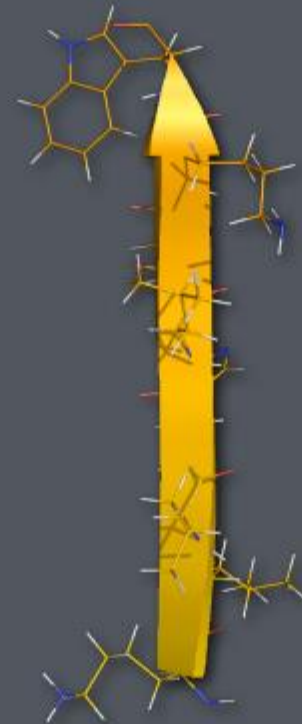
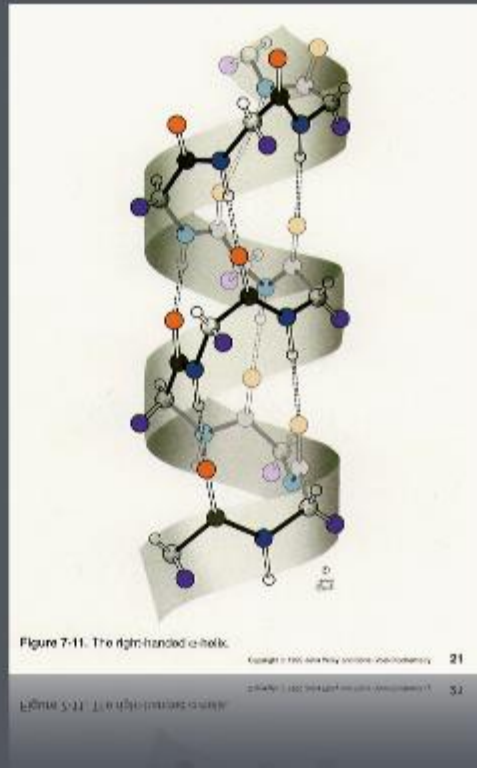
Tighter wound, rare



π helix

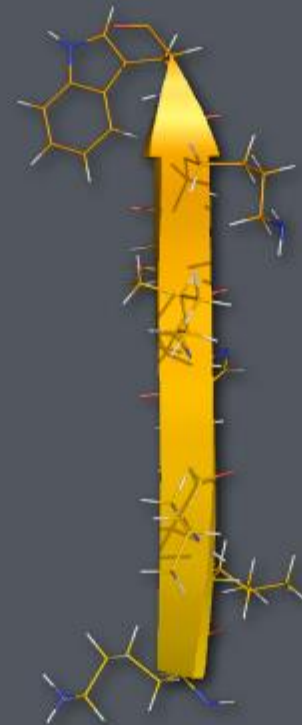
Extremely rare

Discussion: Which secondary structure element is more stable?

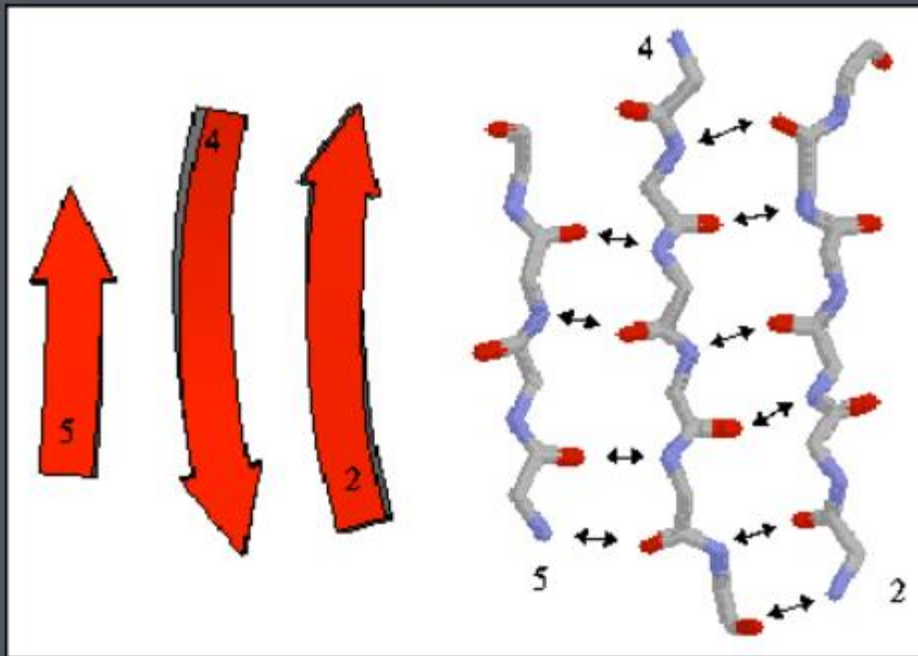


Secondary structure

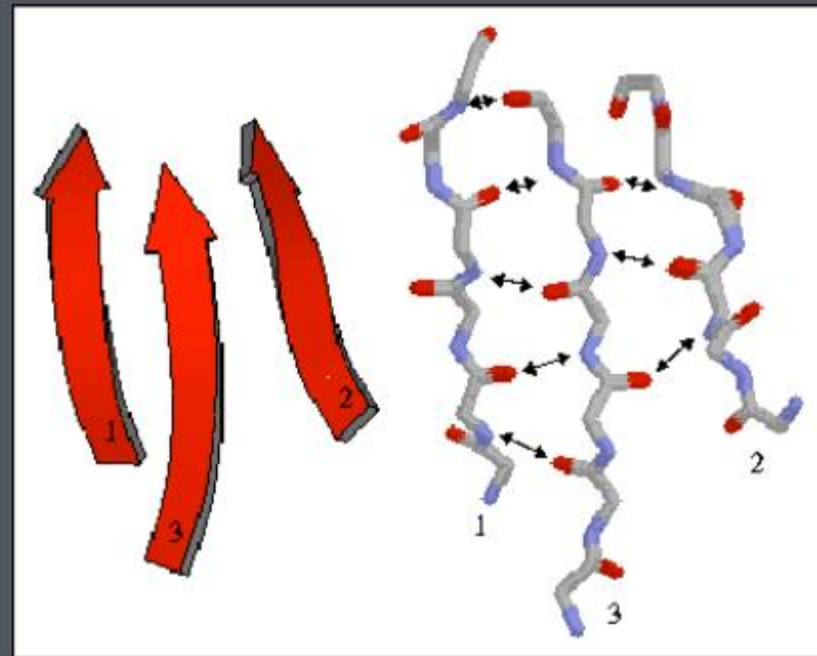
- Local structure is very ordered
 - Helices
 - Sheets
 - Turns
- Stable building blocks
 - Paired hydrogen bonds
 - Good local packing
 - No interference of side chains



Beta sheets



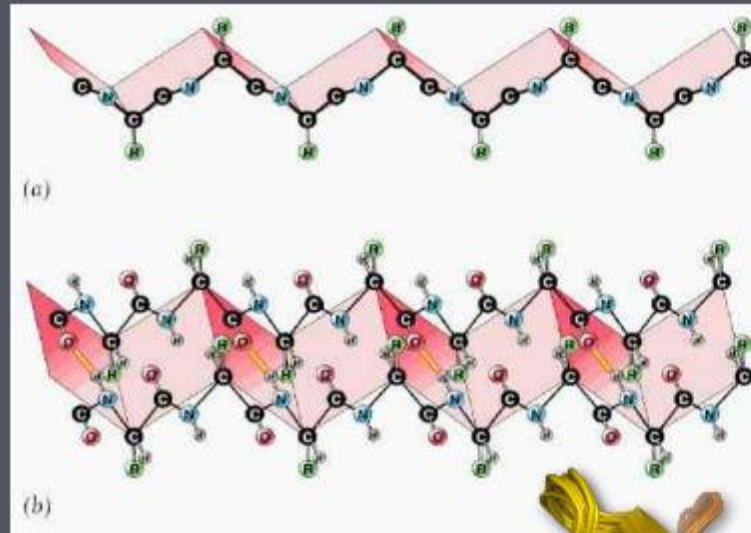
Antiparallel sheets



Parallel sheets

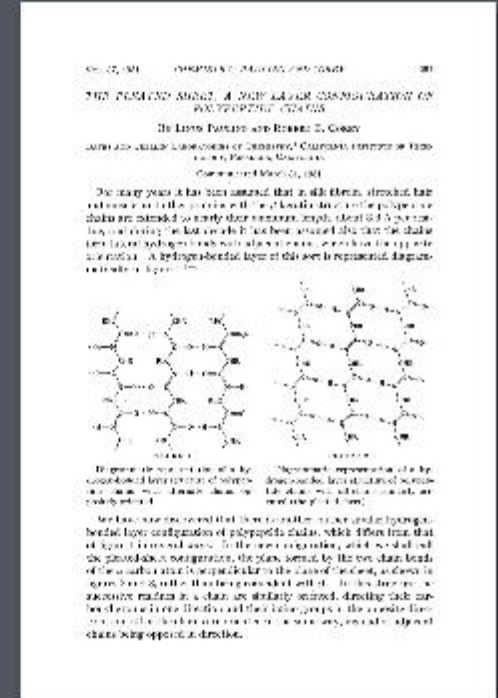
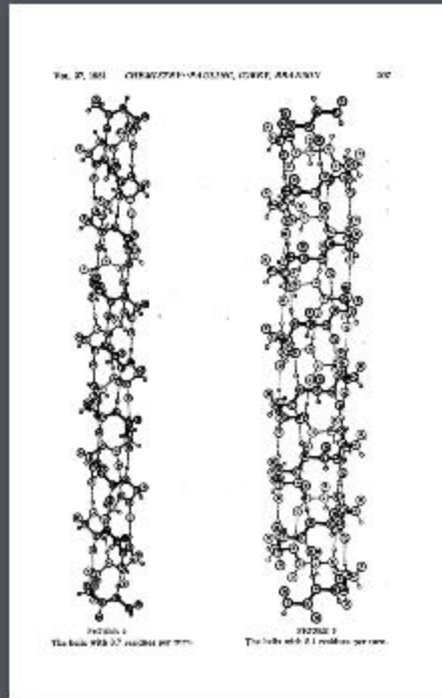
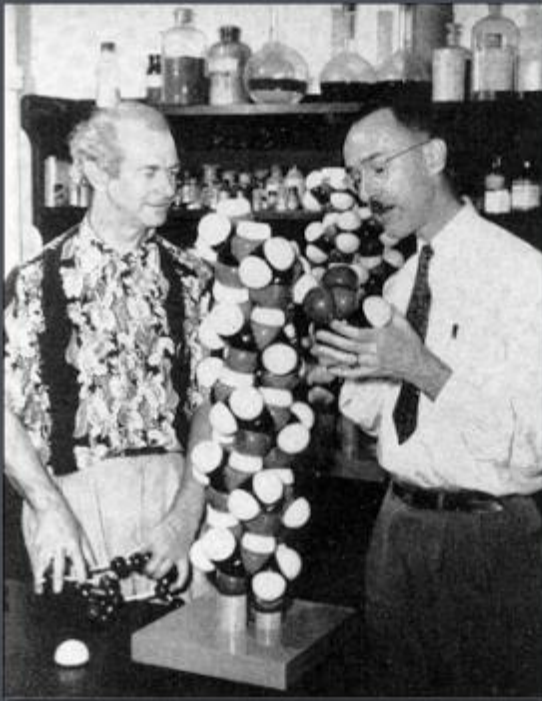
Beta sheets

- Extended chains
- H-bonds between, not inside individual chains
- Pleated sheets
- Slightly twisted, both the individual strand and the whole sheet

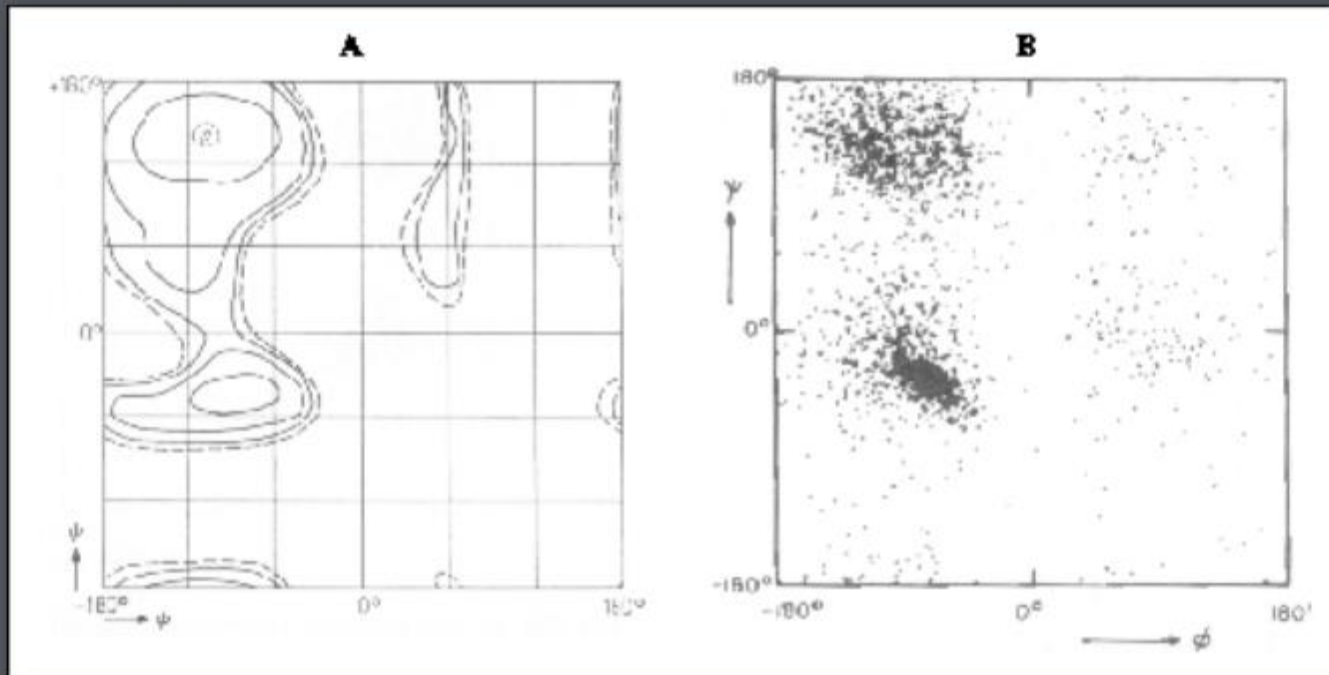


Helix & Sheet discovery

- Pauling, Corey (and partly Branson) - 1951
- “The protein papers” (8 papers in PNAS vol 37)
- Review by Dave Eisenberg (available on Canvas)



Ramachandran, again



- α -helices occupy favorable part of diagram
- 3.6 residues per turn (100 degrees per residue)